

Supplementary Materials for Interaction-aware Transformer with Mixture of Block Attention for protein-ligand docking and affinity prediction (InterMoBA)

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Guide to Supplementary Materials

This document provides additional formulas, tables, and figures that support the main text of our APBC 2026 paper “Interaction-aware Transformer with Mixture of Block Attention for protein-ligand docking and affinity prediction” (InterMoBA). The following items are referenced in the main manuscript:

- **Supplementary Eq. S1:** Detailed calculation of the dynamic chunk size in the MoBA mechanism (referenced in the *MoBA Mechanism* subsection).
- **Supplementary Table S1:** Full results of the MoBA module insertion strategy ablation study (referenced in the *Insertion Strategies of the MoBA Module* subsection).
- **Supplementary Table S2:** Theoretical implications of the Top-K selection ratio (referenced in the *Discussion* section).
- **Supplementary Note 1:** Relationship between sparse attention and PoseBusters PB-valid gap.
- **Supplementary Note 2:** Causes and potential remedies for atomic distance violations.

- **Supplementary Figures S1–S15:** Additional analyses, including ligand flexibility correlation, ranking performance, affinity correlation, feature distribution changes, homology comparison, PoseBusters benchmark, and extensive feature space visualizations (t-SNE, PCA, UMAP, Isomap). These figures extend the analyses presented in the main *Results* section.
- **Supplementary Figure S16:** Detailed illustration of the MoBA chunk and Top-K strategy.

1 Supplementary Methods

1.1 Detailed MoBA Mechanism

1.1.1 Dynamic Partition and Top-K Selection Details

The MoBA mechanism dynamically allocates attention resources by computing an importance score for each node/edge and adjusting feature chunking via a Top-K selection strategy. The chunk size for each region is dynamically set based on an aggregated score of its node and edge features, distance, contact frequency, and centrality.

Supplementary Eq. S1: Calculation of Dynamic Chunk Size. The chunk size for a batch region b is determined as follows:

$$\text{ChunkSize}_b = \max(\text{MergedLen}_b, \text{BaseChunkSize}) \quad (1)$$

where $\text{BaseChunkSize} = 32$ is the minimum granularity used in our experiments. MergedLen_b is the length of the merged key region after fusing importance scores. The fusion score for a node i is calculated as:

$$\text{FusionScore}_i = \text{MLP}_{\text{fuse}}([\text{Imp}_i, \text{ContactFreq}_i, \text{Centrality}_i]) \quad (2)$$

Here, Imp_i is the node importance score, ContactFreq_i is the contact frequency, and Centrality_i is a centrality measure. These scores are then normalized using softmax:

$$\text{SoftmaxScore}_i = \frac{\exp(\text{FusionScore}_i)}{\sum_j \exp(\text{FusionScore}_j)} \quad (3)$$

The Top-K highest-scoring regions are selected for full-precision attention:

$$\text{Top-K} = \text{argsort}(\text{SoftmaxScore})[:k] \quad (4)$$

where $k = \max(10, \lceil 0.05 \times N \rceil)$, and N is the total number of nodes in the input graph.

1.2 Inter-Blocks Computation Details

In the main paper, the Inter-Blocks module is introduced to model cross-molecular interactions between protein and ligand atoms. Due to space constraints, the complete derivation of the attention update was omitted. Here we provide the full formulation that corresponds to Equation (4) in the main text.

Given the updated node features H from the previous Intra-Blocks (Equation (5) in the main paper) and the edge feature transformation E' , the Inter-Blocks apply masked self-attention with a protein–ligand specific mask M'' :

$$H^{\text{inter}} = \text{MaskAttn}(H, E', M''), \quad (5)$$

where MaskAttn is defined in Equation (3) of the main paper. The mask M'' is defined element-wise as:

$$M''_{ij} = \begin{cases} 1 & \text{if } v_i \in V_L, v_j \in V_P \quad (\text{ligand-to-protein direction}), \\ 1 & \text{if } v_i \in V_P, v_j \in V_L \quad (\text{protein-to-ligand direction}), \\ -\infty & \text{otherwise.} \end{cases} \quad (6)$$

This ensures that attention is only computed between ligand and protein nodes, fully capturing bidirectional intermolecular interactions while ignoring intra-molecular pairs.

The edge feature transformation E' is obtained by combining the original distance-based bias A' with a residual update:

$$E' = A' + \text{FFN}(\text{LN}(d)), \quad (7)$$

where A' is the distance-based attention bias (derived from interatomic distances), d denotes the pairwise distance matrix, LN is layer normalization, and FFN is a two-layer feed-forward network. This formulation corresponds exactly to the missing Equation (4) in the main paper and completes the description of the Inter-Blocks module.

1.3 Training Hyperparameters

The model was trained using the AdamW optimizer with an initial learning rate of 1×10^{-4} , a weight decay of 1×10^{-2} , and a batch size of 32. We employed a cosine annealing learning rate scheduler with a warm-up period of 5000 steps. Mixed-precision training (fp16) was used to accelerate computation and reduce memory footprint. Models were trained on a single NVIDIA RTX 3080 GPU for a maximum of 200 epochs, with early stopping based on the validation loss.

1.4 Dataset Split Details

For the time-split PDBBind benchmark, we followed the protocol established in previous works (e.g., DiffDock, Interformer). Specifically, we used the PDBBind v2016 dataset and split complexes based on their PDB release year, ensuring that all test set complexes have a release year later than 2016, while training and validation sets contain older entries. This yields 333 complexes in the test set. The full (random) split is also used for ablation studies.

1.5 Monte Carlo Sampling Procedure

InterMoBA generates candidate ligand poses via a Monte Carlo (MC) sampling algorithm. Starting from an initial random orientation and torsion angles of the ligand within the binding pocket, the MC simulation performs 5000 steps. At each step, a random translation, rotation, or torsion change is proposed, and the energy (negative log-likelihood of the mixture density function) is evaluated. The proposal is accepted according to the Metropolis criterion. The lowest-energy conformations among the sampled trajectory are retained as the Top-K poses (K=10 by default).

2 Supplementary Results

2.1 MoBA Insertion Strategy Analysis

Supplementary Table S1: Performance of InterMoBA variants under Full and Time-Split settings across different MoBA insertion strategies. This table provides the complete numerical results for the MoBA insertion frequency ablation study, corresponding to the analysis in the main text.

Table 1. Performance of InterMoBA variants under Full and Time-Split settings across three different RMSD metrics.

Model	Top-1 RMSD				Top-5 RMSD				Top-10 RMSD			
	Full		Time-Split		Full		Time-Split		Full		Time-Split	
	%<2Å [†]	Median [‡]	%<2Å [†]	Median [‡]	%<2Å [†]	Median [‡]	%<2Å [†]	Median [‡]	%<2Å [†]	Median [‡]	%<2Å [†]	Median [‡]
InterMoBA(Layer1)	68.3	1.07	61.9	1.36	83.4	0.77	77.2	0.96	85.5	0.75	79.0	0.91
InterMoBA(Layer1&2)	67.6	1.15	59.5	1.45	81.1	0.96	74.2	1.10	83.1	0.92	76.6	1.08
InterMoBA(Layer%2)	66.0	1.12	59.4	1.49	81.2	0.93	73.9	1.13	82.7	0.90	75.8	1.05
InterMoBA(Layer%3)	67.7	1.15	59.5	1.54	80.4	0.96	74.3	1.09	82.7	0.93	76.4	1.03
InterMoBA	69.4	1.02	65.5	1.33	83.1	0.86	78.4	1.00	85.2	0.84	79.9	0.96

This table shows the results of retraining the energy task in the InterMoBA model and evaluating it on the time-split evaluation set after the MoBA module is inserted with different invocation strategies (Layer1, Layer1&2, Layer%2, Layer%3, and global insertion). Bold numbers indicate the best performance for each metric.

2.2 Analysis of Top-K Selection Ratio on RMSD and Interaction Recovery

Supplementary Table S2: Theoretical implications of the current Top-K selection ratio.

Table 2. Theoretical implications of the current Top-K selection ratio.

Parameter	Value / Implication
Selection ratio k/N	$\approx 5\%$ (from $k = \max(10, 0.05N)$)
Fraction processed with reduced precision	$> 95\%$
Observed H-bond recovery	63.74% (high)
Observed hydrophobic recovery	54.69% (high)
Observed Top-1 RMSD gain over Interformer	+1.6 percentage points (modest)
Observed PB-valid gap	Visible (Supp. Fig. 6), consistent with limited global information

Since the selection ratio is fixed in the current implementation, we provide here a theoretical analysis of its implications. The 5% retention ratio implies that over 95% of protein-ligand atom pairs are processed with reduced attention precision. While this strategy effectively focuses computational resources on regions with high importance scores, as reflected in the high interaction recovery rates for hydrogen bonds (63.74%) and hydrophobic contacts (54.69%), it also inherently limits the propagation of global structural information. This limitation is consistent with the observation that the Top-1 RMSD success rate (65.5%) exceeds Interformer (63.9%) by only 1.6 percentage points, a modest gain compared to the improvement in interaction recovery.

Supplementary Note 1: Relationship between sparse attention and PoseBusters PB-valid gap.

The atomic distance violations reported in Supplementary Figure 6 provide indirect evidence of the global information limitation. Specifically, the absence of long range constraints in the attention mechanism allows the model to produce poses that satisfy local interaction patterns but may violate steric boundaries between non adjacent atoms. A quantitative sensitivity analysis of the selection ratio, for example varying k from 1% to 20% of N , would be required to establish a causal relationship. We leave this investigation for future work.

2.3 Analysis of Physical Plausibility and PoseBusters Results

Supplementary Figure 6 (expanded caption): The figure compares RMSD based success rates (dark bars) with PB-valid rates (light bars) for InterMoBA and baseline methods. The gap between the two bars indicates the fraction of poses that achieve RMSD $< 2 \text{ \AA}$ but violate at least one PoseBusters validity criterion. According to the PoseBusters report, the most frequent

violation for deep learning based docking methods is atomic distance clashes between ligand and protein atoms. InterMoBA exhibits a similar pattern, suggesting that the MDN based energy function, which is trained to maximize the likelihood of observed interatomic distances without a penalty for van der Waals overlaps, does not inherently prevent steric clashes.

Supplementary Note 2: Causes and potential remedies for atomic distance violations.

Causes:

1. The training loss (Equation 11 in the main text) maximizes the conditional probability density of distances but does not include a repulsive term for distances below the sum of van der Waals radii.
2. The Monte Carlo sampling procedure (5000 steps) uses the negative log likelihood as the energy function; without a clash penalty, low energy poses may still contain penetrations.
3. The Top-K sparse attention reduces information exchange across distant atom pairs, which may weaken the model’s ability to enforce global steric constraints.

Potential remedies (not implemented in this work):

1. Augment the MDN loss with a steric clash penalty term: $L_{\text{total}} = L_{\text{MDN}} + \lambda \sum_{i,j} \max(0, r_{\text{vdW}} - d_{ij})^2$.
2. Increase the Top-K selection ratio, for example to 10% or 20% of N , to improve global information flow at the cost of computational efficiency.
3. Apply post hoc energy minimization using a classical force field such as MMFF94 to resolve clashes while preserving the interaction pattern.

3 Supplementary Figures

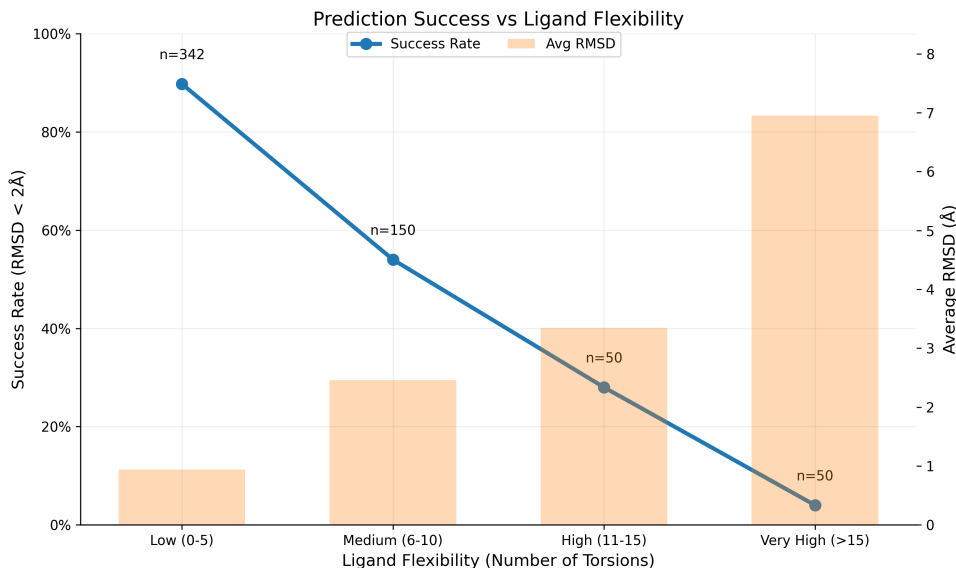


Figure 1. Correlation analysis between ligand flexibility and prediction success rate. Increasing ligand flexibility (rotatable bond count) significantly reduces prediction success rate (RMSD < 2 Å), from ~90% (low-flex) to <10% (high-flex), with rising RMSD, highlighting flexibility as a critical accuracy bottleneck.

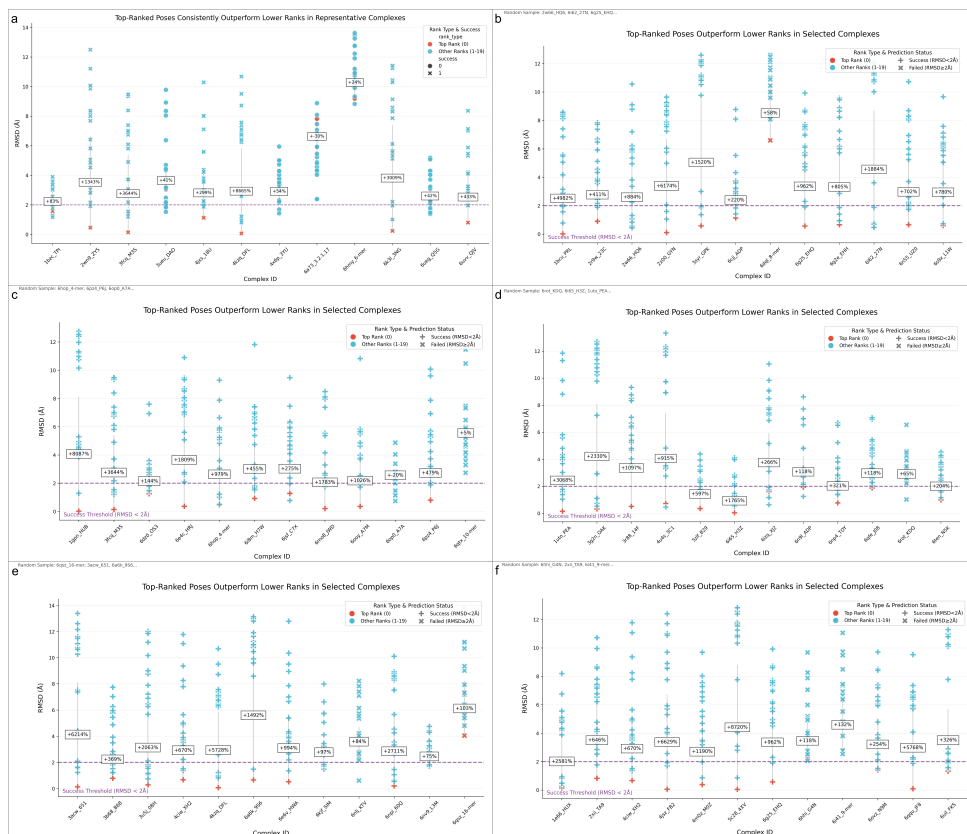


Figure 2. Effectiveness of top-ranked poses. The six sub-diagrams collectively illustrate that top-ranked poses consistently exhibit lower RMSD values across various complexes, underscoring the model’s robustness in ranking poses.

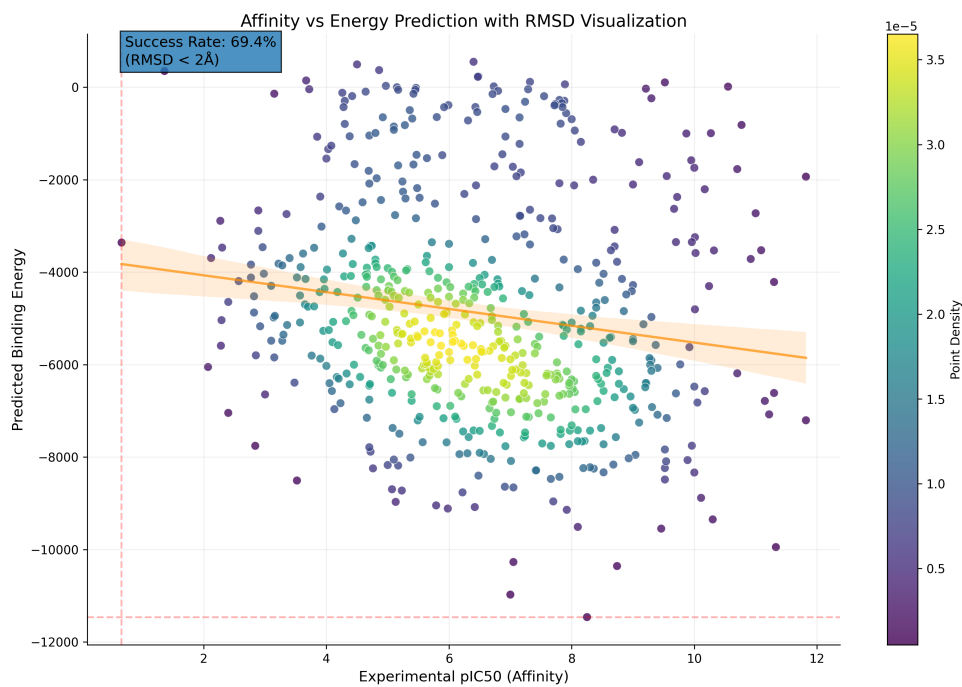


Figure 3. Correlation between experimental pIC50 and predicted binding energy. A negative correlation is observed, demonstrating the model’s dual capability in affinity discrimination and conformational prediction.

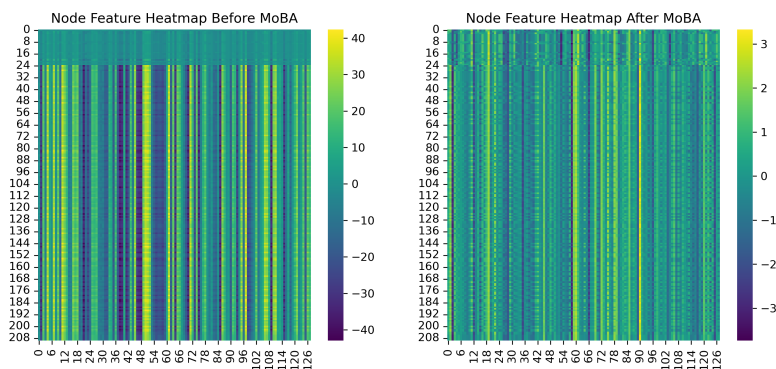


Figure 4. Impact of MoBA operation on node feature distribution. After MoBA, feature values become more concentrated and evenly distributed, indicating effective normalization and adjustment.

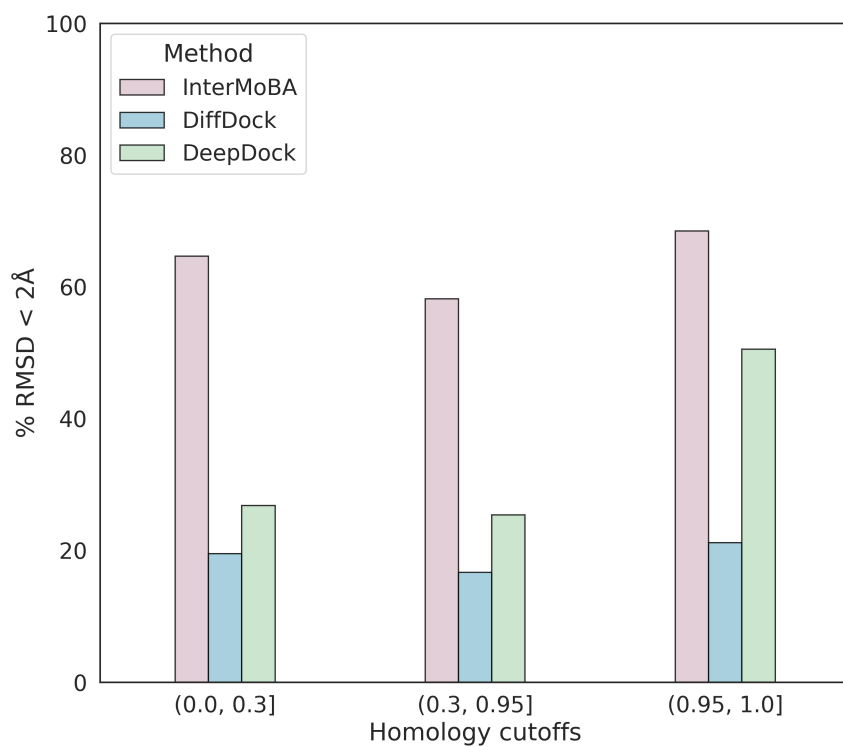


Figure 5. Docking accuracy across homology subsets. InterMoBA consistently outperforms DiffDock and DeepDock across low (0.0-0.3), medium (0.3-0.95), and high (0.95-1.0) homology subsets. Accuracy is measured as the percentage of poses with $\text{RMSD} < 2 \text{ \AA}$.

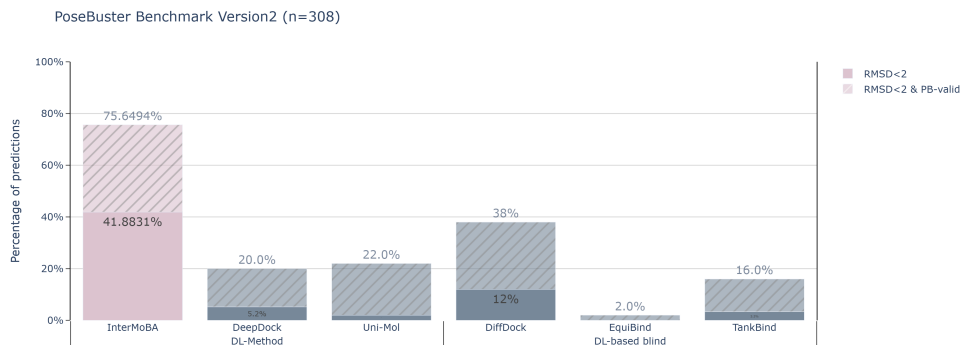


Figure 6. Comparison of success rates on the PoseBusters Benchmark Version 2. The dark section represents predictions with $\text{RMSD} < 2 \text{ \AA}$; the light section indicates poses that additionally pass PoseBusters verification (PB-valid). InterMoBA shows strong RMSD performance but has room for improvement in physical plausibility checks, primarily due to atomic distance violations.

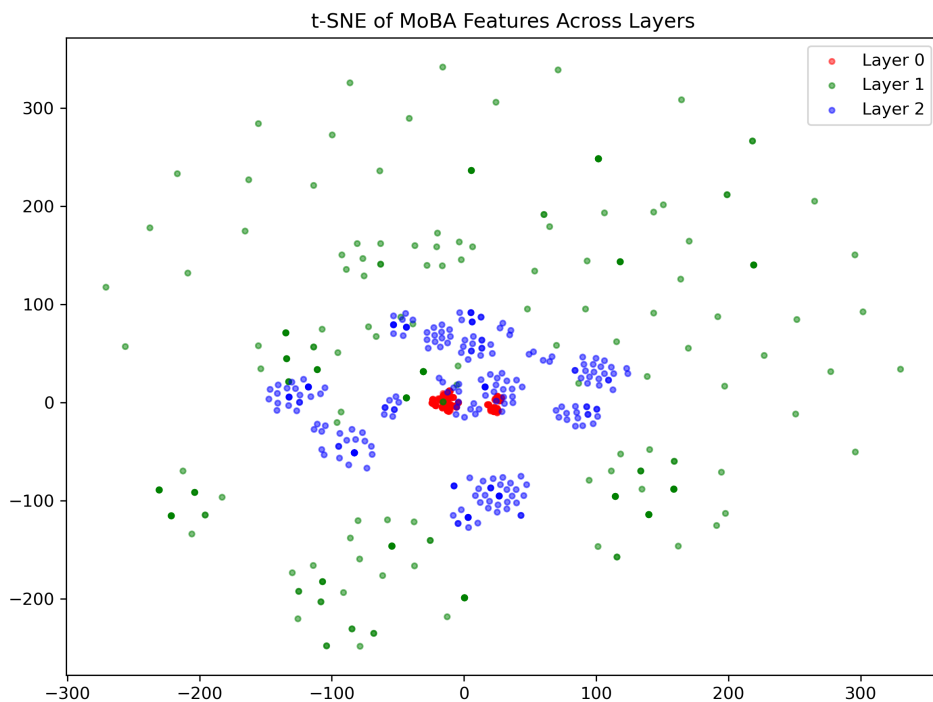


Figure 7. t-SNE visualization of inter node features across layers. The 2D projection shows how feature clustering and spatial distribution evolve across layers 0, 1, and 2 of the MoBA model, demonstrating improved separation after MoBA.

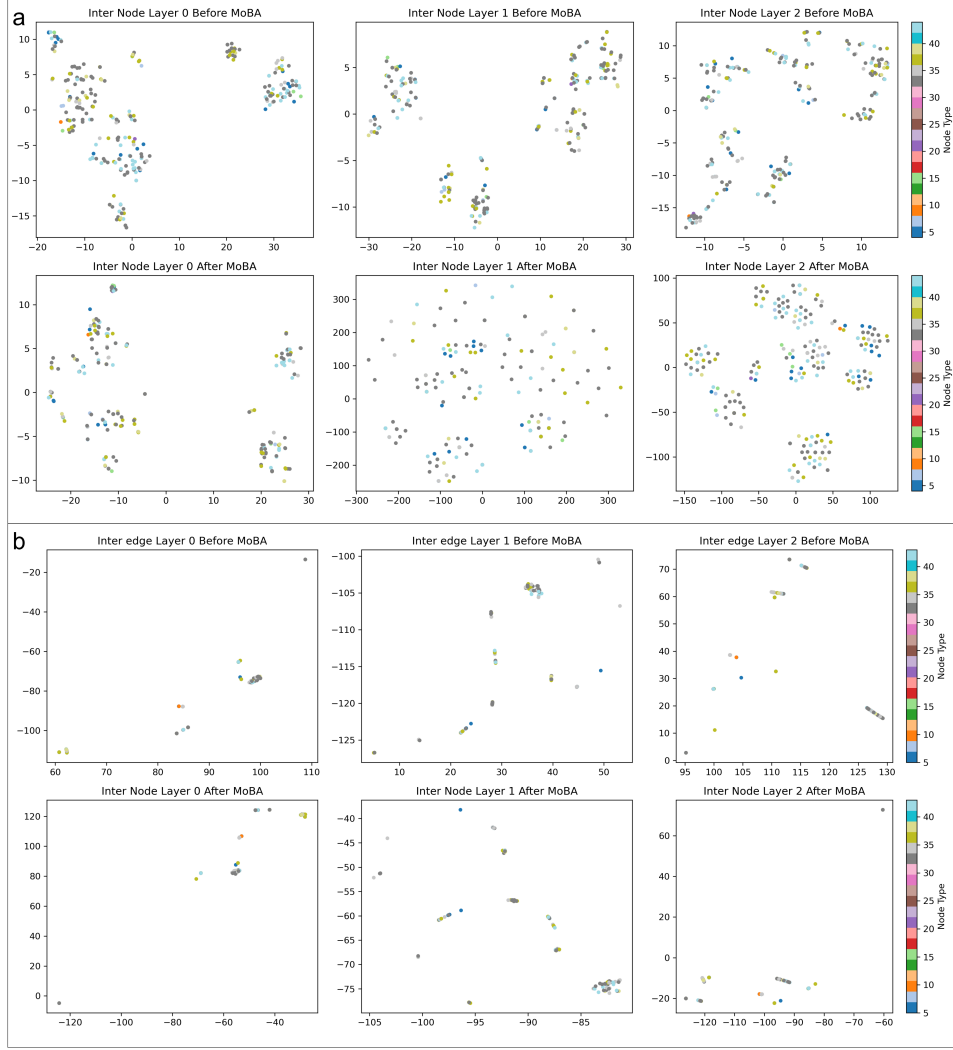


Figure 8. t-SNE visualizations of node and edge features before and after MoBA. (a) Node features; (b) Edge features. Colors indicate node types. MoBA enhances the structural organization of both node and edge feature spaces.

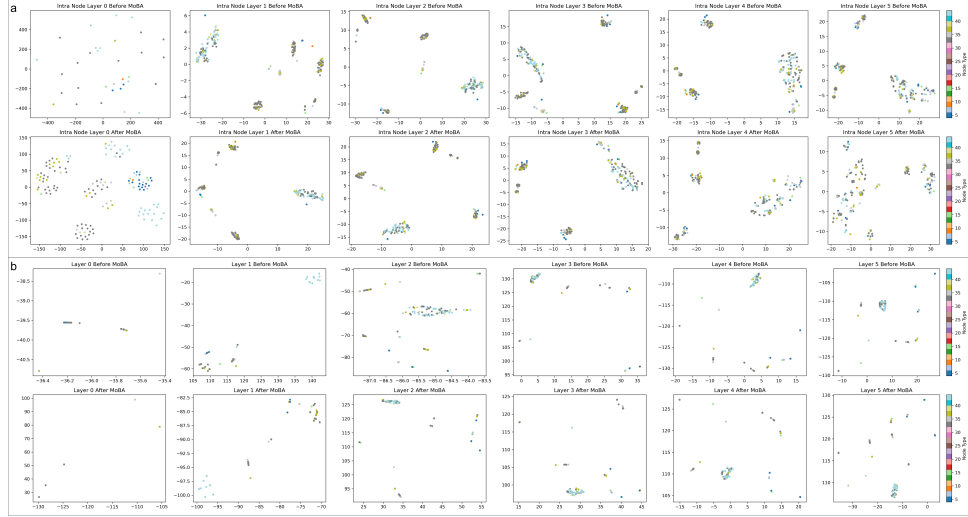


Figure 9. t-SNE visualizations of intra node and intra edge features across layers. (a) Intra node features; (b) Intra edge features. MoBA processing leads to improved clustering and distribution differences at each layer.

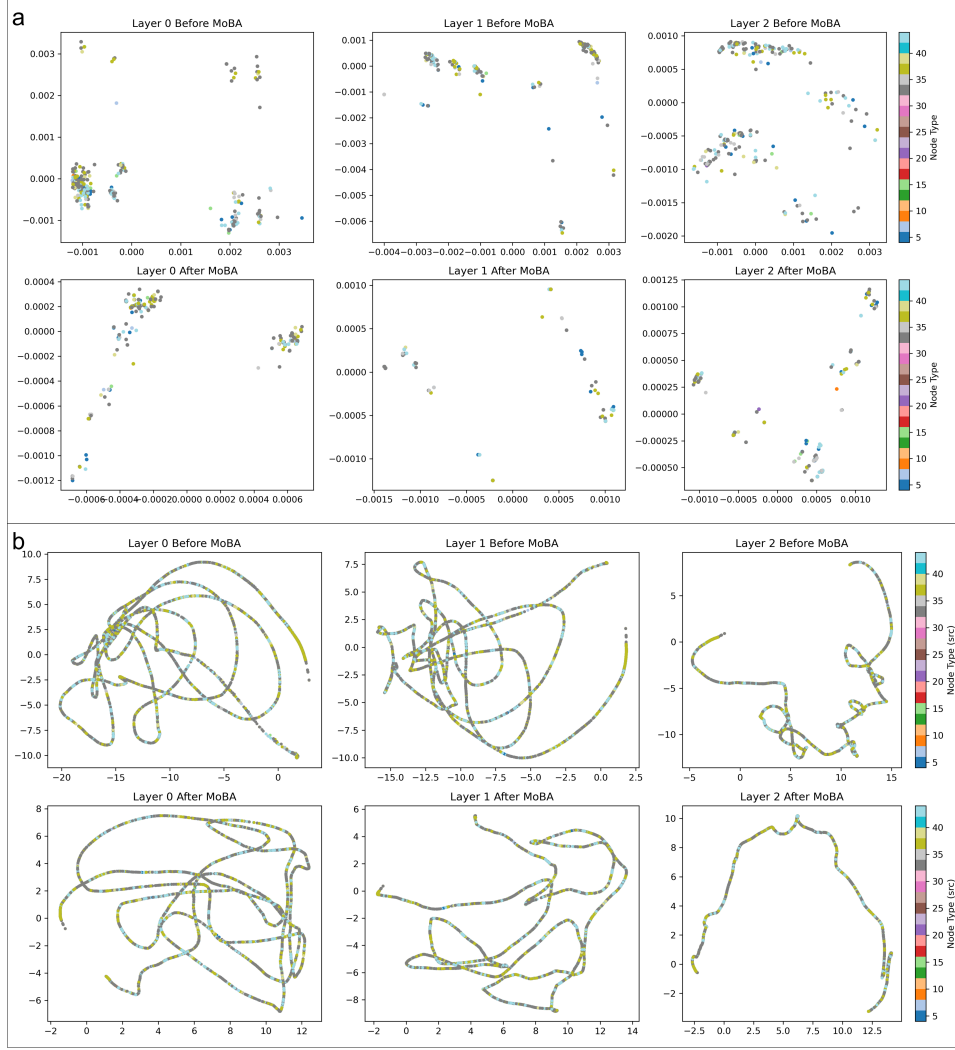


Figure 10. PCA-reduced distributions of inter node and inter edge features. (a) Inter node features; **(b)** Inter edge features. Colors denote node types, showing the impact of MoBA on feature distribution and type separability.

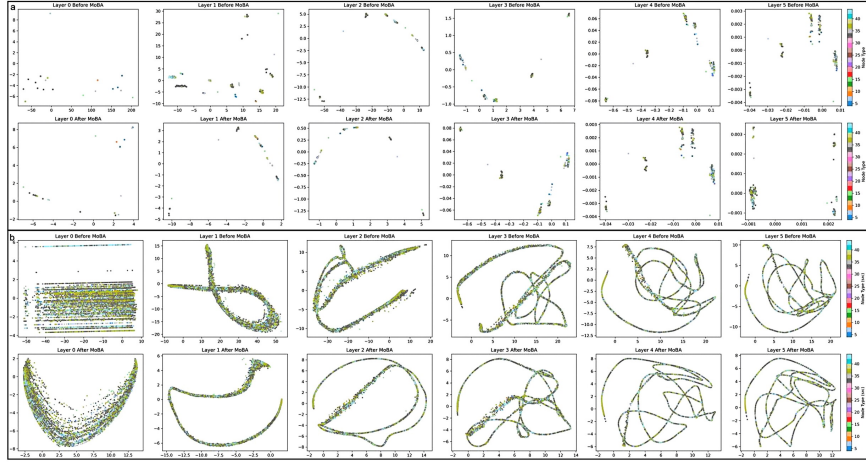


Figure 11. PCA-reduced distributions of intra node and intra edge features. (a) Intra node features; (b) Intra edge features. MoBA improves feature distribution, type separability, and edge feature organization across layers.

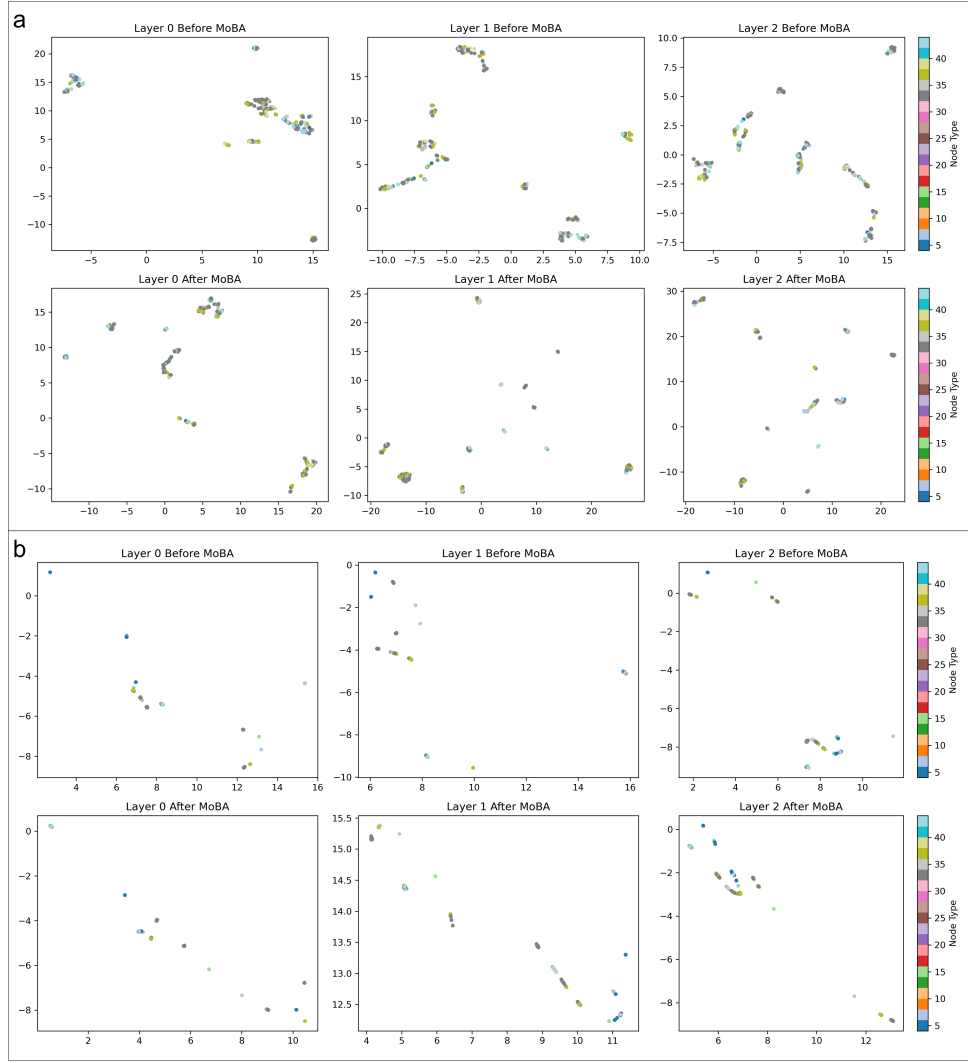


Figure 12. UMAP projections of inter node and inter edge features. (a) Inter node features; (b) Inter edge features. UMAP confirms that MoBA enhances clustering structure, with colors indicating node types.

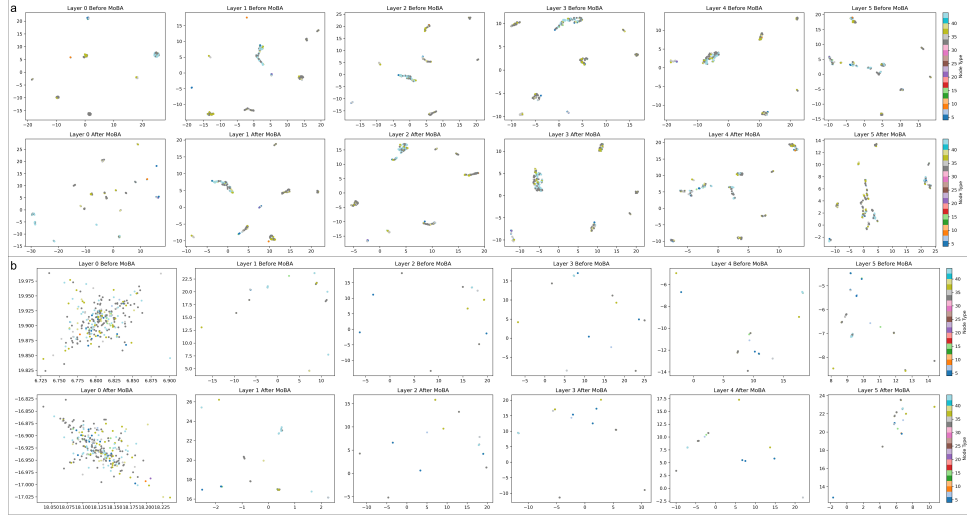


Figure 13. UMAP projections of intra node and intra edge features. (a) Intra node features; (b) Intra edge features. MoBA improves the clustering structure of node and edge features.

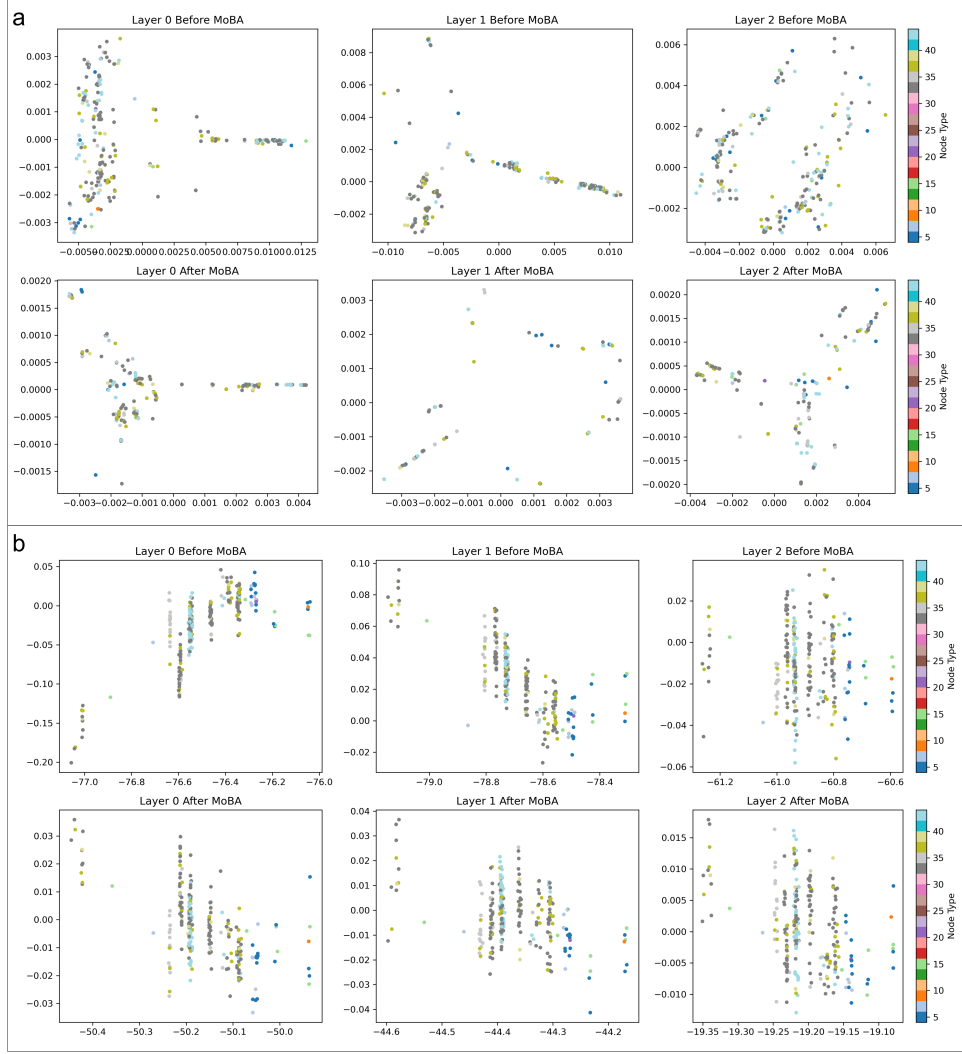


Figure 14. Isomap projections of inter node and inter edge features. (a) Inter node features; (b) Inter edge features. Isomap reveals changes in feature space structure due to MoBA, with colors indicating node types.

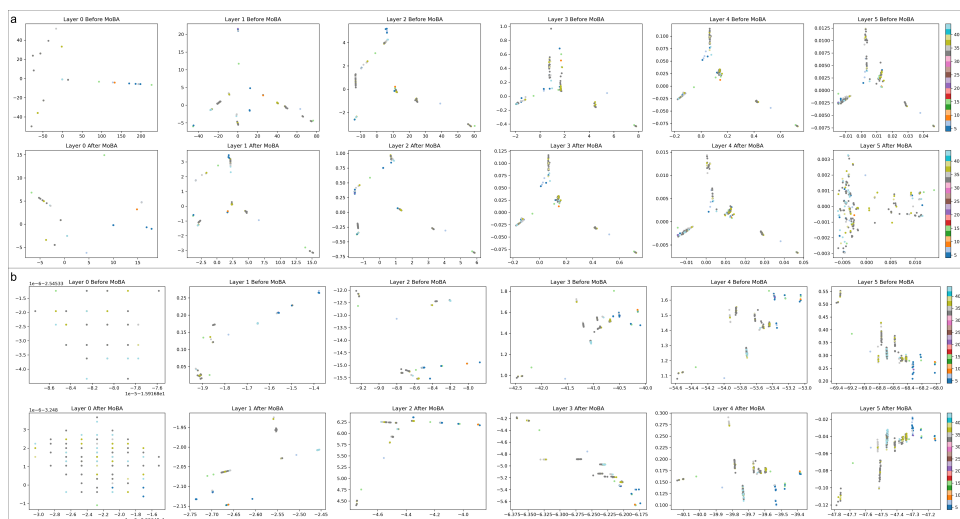


Figure 15. Isomap projections of intra node and intra edge features. (a) Intra node features; (b) Intra edge features. MoBA induces clear changes in feature space structure, improving clustering and distribution.

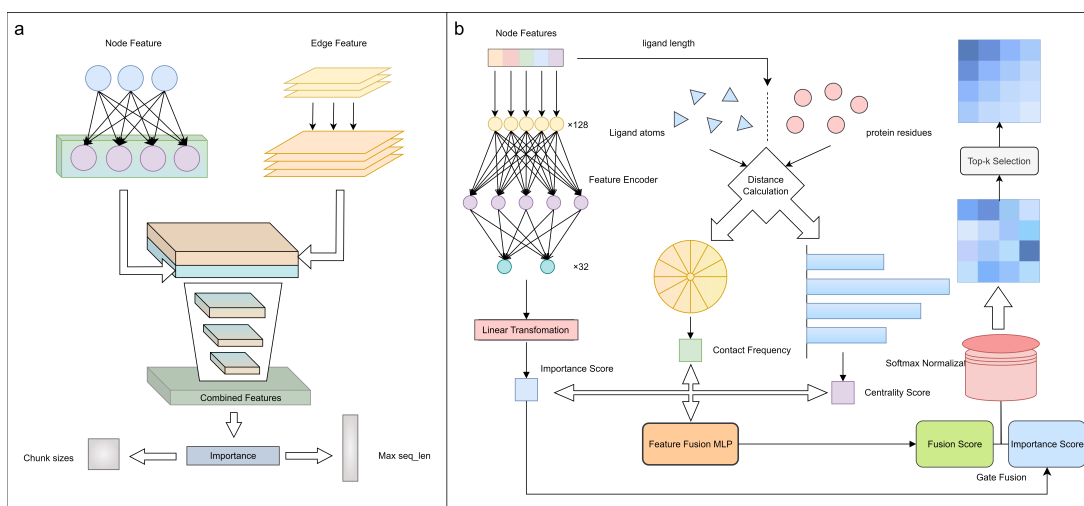


Figure 16. Feature-fusion-based chunk and Top-K strategy.

a Chunk strategy. Node and edge features are extracted separately, fused via a learnable projection, and used to compute per-residue importance scores. Chunk boundaries are then adjusted dynamically according to these scores and spatial proximity, while respecting a maximum sequence-length limit, enabling efficient and context-aware processing.

b Top-K strategy. Multiple signals (feature encoding, inter-node distances, and contact frequency) are fused to yield a normalized importance score for each node. The Top-K highest-scoring nodes are retained for downstream analysis and modeling, substantially improving computational efficiency and predictive accuracy.